

# Supplementary Material for “Toward a Tractability Frontier for Exact Relevance Certification”

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## 1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

| ID   | Lean Handle / Source                                                              | ID   | Lean Handle / Source                                                                            |
|------|-----------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------|
| FR1  | currentFamilies_partition_by_role<br>Paper4dFrontier/Classification.lean          | FR2  | currentFamilies_partition_counts<br>Paper4dFrontier/Classification.lean                         |
| FR3  | current_families_have_finite_basis<br>Paper4dFrontier/Classification.lean         | FR4  | current_positive_interfaces_factor_through_role<br>Paper4dFrontier/Classification.lean          |
| FR5  | lifted_families_reduce_to_core<br>Paper4dFrontier/Classification.lean             | FR6  | degenerateFamilies_complete<br>Paper4dFrontier/Classification.lean                              |
| FR7  | exists_decisionProblem_realizing_labeling<br>Paper4dFrontier/Realizability.lean   | FR8  | realizingProblem_decisionEquiv_iff<br>Paper4dFrontier/Realizability.lean                        |
| FR13 | isOptimal_positiveAffineTransform_iff<br>Paper4dFrontier/FamilyAxioms.lean        | FR14 | opt_eq_positiveAffineTransform<br>Paper4dFrontier/FamilyAxioms.lean                             |
| FR15 | isSufficient_positiveAffineTransform_iff<br>Paper4dFrontier/FamilyAxioms.lean     | FR16 | isRelevant_positiveAffineTransform_iff<br>Paper4dFrontier/FamilyAxioms.lean                     |
| FR17 | decisionEquiv_relabelActions_iff<br>Paper4dFrontier/FamilyAxioms.lean             | FR18 | decisionEquiv_relabelStates_iff<br>Paper4dFrontier/FamilyAxioms.lean                            |
| FR19 | isSufficient_relabelActions_iff<br>Paper4dFrontier/FamilyAxioms.lean              | FR20 | isSufficient_relabelStates_iff<br>Paper4dFrontier/FamilyAxioms.lean                             |
| FR21 | allProblems_relabelInvariant<br>Paper4dFrontier/FamilyAxioms.lean                 | FR22 | allProblems_kernelUniversal<br>Paper4dFrontier/FamilyAxioms.lean                                |
| FR23 | relabelInvariant_not_enough<br>Paper4dFrontier/FamilyAxioms.lean                  | FR24 | optRangeCard_realizingIdentity<br>Paper4dFrontier/FamilyAxioms.lean                             |
| FR25 | currentFamilies_explicit<br>Paper4dFrontier/Classification.lean                   | FR26 | coreFamilies_explicit<br>Paper4dFrontier/Classification.lean                                    |
| FR27 | liftedFamilies_explicit<br>Paper4dFrontier/Classification.lean                    | FR28 | degenerateFamilies_explicit<br>Paper4dFrontier/Classification.lean                              |
| FR29 | primitiveMechanisms_explicit<br>Paper4dFrontier/Classification.lean               | FR30 | isSufficient_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                      |
| FR31 | isRelevant_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean          | FR32 | isIrrelevant_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                      |
| FR33 | isMinimalSufficient_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean | FR34 | sufficientSets_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                     |
| FR35 | relevantSet_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean          | FR37 | quotientCard_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                       |
| FR38 | quotientCard_realizingIdentity<br>Paper4dFrontier/FamilyAxioms.lean               | FR39 | realizingProblemQuotientEquivLabelRange_apply_quotientMap<br>Paper4dFrontier/Realizability.lean |

(continued...)

| ID    | Lean Handle / Source                                                                                              | ID    | Lean Handle / Source                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------|
| FR40  | realizingProblem_quotientCard_eq_labelRangeCard<br>Paper4dFrontier/Realizability.lean                             | FR41  | certificationStatistic_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                              |
| FR42  | sufficientSetCount_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                                   | FR43  | minimalSufficientSetCount_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                           |
| FR44  | relevantCoordCount_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                                   | FR45  | setoidRealizingProblem_decisionEquiv_iff<br>Paper4dFrontier/Realizability.lean                                   |
| FR46  | setoidRealizingProblemQuotientEquivSetoidQuotient_apply_quotientMap<br>Paper4dFrontier/Realizability.lean         | FR47  | exists_decisionProblem_realizing_setoid<br>Paper4dFrontier/Realizability.lean                                    |
| FR48  | isSufficient_iff_agreementRel_le_decisionEquiv<br>Paper4dFrontier/FamilyAxioms.lean                               | FR49  | isIrrelevant_iff_sufficient_erase<br>Paper4dFrontier/FamilyAxioms.lean                                           |
| FR50  | isRelevant_iff_not_sufficient_erase<br>Paper4dFrontier/FamilyAxioms.lean                                          | FR51  | quotientEquiv_of_decisionEquiv_iff_apply_quotientMap<br>Paper4dFrontier/FamilyAxioms.lean                        |
| FR52  | empty_sufficient_of_constant_opt<br>Paper4dFrontier/ClosureLaws.lean                                              | FR53  | irrelevant_of_constant_opt<br>Paper4dFrontier/ClosureLaws.lean                                                   |
| FR54  | decisionEquiv_duplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                             | FR55  | isSufficient_duplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                             |
| FR56  | isRelevant_duplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                                | FR57  | decisionEquiv_duplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                             |
| FR58  | isSufficient_duplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                               | FR59  | isRelevant_duplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                                |
| FR60  | duplicateStateQuotientEquivOriginal_apply_quotientMap<br>Paper4dFrontier/ClosureLaws.lean                         | FR66  | booleanCube_card<br>Paper4dFrontier/EnumerationBounds.lean                                                       |
| FR67  | exists_booleanCube_larger_than_monomial<br>Paper4dFrontier/EnumerationBounds.lean                                 | FR68  | decisionEquiv_addDuplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                          |
| FR69  | isSufficient_addDuplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                            | FR70  | isRelevant_addDuplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                             |
| FR71  | addDuplicateStateQuotientEquivOriginal_apply_quotientMap<br>Paper4dFrontier/ClosureLaws.lean                      | FR72  | some_mem_opt_addDuplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                          |
| FR73  | none_mem_opt_addDuplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                           | FR83  | isSufficient_noiseExtended_iff<br>Paper4dFrontier/DimensionalNoiseExtension.lean                                 |
| FR84  | isRelevant_noiseExtended_iff<br>Paper4dFrontier/DimensionalNoiseExtension.lean                                    | FR85  | lastCoord_irrelevant_noiseExtended<br>Paper4dFrontier/DimensionalNoiseExtension.lean                             |
| FR100 | parentTree_realTreewidth_one<br>Paper4dFrontier/ParentTreewidth.lean                                              | FR101 | low_rank_only_witness<br>Paper4dFrontier/LowRankOnlyWitness.lean                                                 |
| FR105 | separable_only_witness<br>Paper4dFrontier/SeparableOnlyWitness.lean                                               | FR106 | bounded_actions_only_witness<br>Paper4dFrontier/BoundedActionsOnlyWitness2.lean                                  |
| FR108 | all_independent_core_mechanisms_nondegenerate<br>Paper4dFrontier/Block2Progress.lean                              | FR110 | parentTreeStructured_implies_treeStructured<br>Paper4dFrontier/ParentTreewidth.lean                              |
| FR111 | treeStructured_and_unique_parent_iff_parentTreeStructured<br>Paper4dFrontier/ParentTreewidth.lean                 | FR112 | bounded_treewidth_only_witness<br>Paper4dFrontier/BoundedTreewidthOnlyWitness.lean                               |
| FR113 | symmetry_only_witness<br>Paper4dFrontier/SymmetryOnlyWitness.lean                                                 | FR114 | weak_tree_structured_not_width_one<br>Paper4dFrontier/LegacyTreeGap.lean                                         |
| FR115 | tree_structure_hardness_bridge<br>Paper4dFrontier/TreeStructureHardnessBridge.lean                                | FR118 | pairCrossDifference_eq_binaryCrossDifference_of_lt<br>Paper4dFrontier/BinaryPairwiseDichotomy.lean               |
| FR119 | pairwise_zero_crossDifference_unaryDecomposition<br>Paper4dFrontier/BinaryPairwiseDichotomy.lean                  | FR120 | binary_pairwise_symmetry_dichotomy<br>Paper4dFrontier/BinaryPairwiseDichotomy.lean                               |
| FR121 | actionGapCrossDifference_eq_binaryCrossDifference_of_lt<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean | FR122 | decisionRelevant_zero_actionGap_implies_unaryReduction<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean |
| FR123 | binary_pairwise_symmetry_decision_relevant_dichotomy<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean    | FR124 | block6_genuine_interaction_dichotomy_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                       |

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| ID    | Lean Handle / Source                                                                                                            | ID    | Lean Handle / Source                                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------|
| FR125 | decisionRelevantInteractionGraph_addActionOffset<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean                      | FR126 | action_offset_can_force_constant_optimizer<br>Paper4dFrontier/Block6Obstruction.lean                                             |
| FR127 | block6_action_offset_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                      | FR128 | offsetNormalizedDecisionRelevantInteractionGraph_wellDefined<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean           |
| FR129 | block6_offset_normalized_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                  | FR130 | neverOptimalGhost_decisionRelevantGraph_eq_top<br>Paper4dFrontier/Block6Obstruction.lean                                         |
| FR131 | neverOptimalGhost_supportedGraph_eq_bot<br>Paper4dFrontier/Block6Obstruction.lean                                               | FR132 | support_filtering_removes_known_block6_obstructions<br>Paper4dFrontier/Block6Obstruction.lean                                    |
| FR133 | block6_ghost_action_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                       | FR134 | marginMasking_supportedGraph_eq_top<br>Paper4dFrontier/Block6Obstruction.lean                                                    |
| FR135 | marginMasking_zero_sufficient<br>Paper4dFrontier/Block6Obstruction.lean                                                         | FR136 | block6_optimizer_supported_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                 |
| FR137 | MarginBounded<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean                                                         | FR138 | dominantPair_marginBounded<br>Paper4dFrontier/Block6Obstruction.lean                                                             |
| FR139 | dominantPair_supportedGraph_eq_top<br>Paper4dFrontier/Block6Obstruction.lean                                                    | FR140 | block6_margin_bounded_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                      |
| FR145 | not_collapseLandscapeInfinity_of_oracle_predicate<br>Paper4dFrontier/MetaCharacterization.lean                                  | FR176 | DecisionQuotient.DecisionProblem.quotient_is_coarsest<br>paper4/DecisionQuotient/Quotient.lean                                   |
| FR177 | DecisionQuotient.DecisionProblem.quotient_has_unique_factorization<br>paper4/DecisionQuotient/Quotient.lean                     | FR178 | DecisionQuotient.DecisionProblem.srank_eq_relevant_card<br>paper4/DecisionQuotient/Tractability/StructuralRank.lean              |
| FR179 | DecisionQuotient.quotientEntropy_le_srank_binary<br>paper4/DecisionQuotient/Information.lean                                    | FR180 | DecisionQuotient.Statistics.fisherMatrix_rank_eq_srank<br>paper4/DecisionQuotient/Statistics/FisherInformation.lean              |
| FR181 | DecisionQuotient.Information.compression_below_srank_fails<br>paper4/DecisionQuotient/Information/RDSrank.lean                  | FR182 | DecisionQuotient.Information.srank_bits_sufficient<br>paper4/DecisionQuotient/Information/RDSrank.lean                           |
| FR183 | DecisionQuotient.Physics.single_future_zero_cost<br>paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean                   | FR184 | DecisionQuotient.Physics.transportCost_pos_of_offDiag<br>paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean               |
| FR185 | no_closureInvariant_predicate_of_orbit_gap<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                               | FR186 | no_admissibleNormalizationPredicate_decides_dominantPair<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                  |
| FR187 | no_admissibleNormalizationPredicate_decides_marginBounded<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                | FR188 | no_admissibleNormalizationPredicate_decides_ghostActionTwoPairCrossOne<br>Paper4dFrontier/ObstructionPredicateCandidates.lean    |
| FR189 | no_admissibleNormalizationPredicate_decides_offsetActionZeroPairCrossOne<br>Paper4dFrontier/ObstructionPredicateCandidates.lean | FR190 | admissibleCollapseLandscapeInfinity_full_paper<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                            |
| FR191 | ClosureSoundPackage<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                                      | FR192 | ReasonableGuardrailPackage<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                                |
| FR193 | no_closureSoundPackage_predicate_decides_four_obstruction_families<br>Paper4dFrontier/ObstructionPredicateCandidates.lean       | FR194 | no_reasonableGuardrailPackage_predicate_decides_four_obstruction_families<br>Paper4dFrontier/ObstructionPredicateCandidates.lean |
| FR197 | classifier_agrees_on_closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/AdmissibleCharacterization.lean                    | FR198 | no_correctOnDomain_classifier_of_orbit_gap<br>Paper4dFrontier/AdmissibleCharacterization.lean                                    |
| FR199 | correct_classifier_inherits_closureLawInvariant<br>Paper4dFrontier/AdmissibleCharacterization.lean                              | FR200 | admissibleNormalizationPredicate_has_explicit_inhabitants<br>Paper4dFrontier/AdmissibleCharacterization.lean                     |
| FR201 | closureLawInvariant_of_iff_of_closureEquivalent<br>Paper4dFrontier/AdmissibleCharacterization.lean                              | FR202 | exists_orbit_gap_of_not_closureLawInvariant<br>Paper4dFrontier/AdmissibleCharacterization.lean                                   |

(continued...)

| ID    | Lean Handle / Source                                                                                                   | ID    | Lean Handle / Source                                                                                                    |
|-------|------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------|
| FR203 | closureLawInvariant_iff_no_orbit_gap<br>Paper4dFrontier/AdmissibleCharacterization.lean                                | FR204 | no_exact_closureLawInvariant_classifier_iff_exists_orbit_gap<br>Paper4dFrontier/AdmissibleCharacterization.lean         |
| FR205 | distinctActionCount<br>Paper4dFrontier/DistinctActionProfiles.lean                                                     | FR206 | actionCount_profileCompressedSlice<br>Paper4dFrontier/DistinctActionProfiles.lean                                       |
| FR207 | decisionEquiv_profileCompressedSlice_iff<br>Paper4dFrontier/DistinctActionProfiles.lean                                | FR208 | profileCompressedSlice_preserves_exactCertification<br>Paper4dFrontier/DistinctActionProfiles.lean                      |
| FR209 | profileCompressedSlice_bounded_actions<br>Paper4dFrontier/DistinctActionProfiles.lean                                  | FR210 | OrbitGapOn<br>Paper4dFrontier/AdmissibleCharacterization.lean                                                           |
| FR211 | no_orbitGapOn_of_exact_classifiable_by_closureLawInvariant_onDomain<br>Paper4dFrontier/AdmissibleCharacterization.lean | FR212 | exact_classifiable_by_closureLawInvariant_onDomain_iff_no_orbitGapOn<br>Paper4dFrontier/AdmissibleCharacterization.lean |
| FR213 | no_exact_closureLawInvariant_classifier_onDomain_iff_orbitGapOn<br>Paper4dFrontier/AdmissibleCharacterization.lean     | FR214 | boundedPatternScheme_holds_largeActionCount_iff<br>Paper4dFrontier/AdmissibleCharacterization.lean                      |
| FR215 | boundedPatternDefinable_eventually_constant_in_actionCount<br>Paper4dFrontier/AdmissibleCharacterization.lean          | FR216 | boundedPatternDefinable_largeActionCount_agrees<br>Paper4dFrontier/AdmissibleCharacterization.lean                      |

## 2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

| Paper claim                                                                         | Lean handle                                                            |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Corollary 6.6: Constant-Optimizer Collapse Forces Trivial Certification             | FR52, FR53                                                             |
| Corollary 2.3: Finite Basis for the Current Positive Landscape                      | FR3, FR29                                                              |
| Corollary 4.11: No Correct Tractability Classifier                                  | <i>(no derived Lean handle found)</i>                                  |
| Corollary 4.13: Orbit-Gap Completeness on Closure-Closed Domains                    | FR210, FR212, FR213, FR211                                             |
| Corollary 5.4: Realizability Alone Cannot Be the Frontier                           | FR7, FR8                                                               |
| Corollary 4.10: Reasonable Guardrail Packages                                       | FR194                                                                  |
| Corollary 6.3: Relevance Is Failure of Erased Sufficiency                           | FR49, FR50                                                             |
| Corollary 6.4: Certification Statistics Factor Through the Quotient Relation        | FR41, FR43, FR44, FR42                                                 |
| Corollary 6.2: Sufficiency Is Relation Refinement                                   | FR48                                                                   |
| Proposition 6.18: The Admissibility Layer Is Nonempty                               | FR200                                                                  |
| Proposition 6.8: Positive Affine Utility Reparameterizations Are Invisible          | FR13, FR16, FR15, FR14                                                 |
| Proposition 4.1: Binary Pairwise Symmetry Dichotomy                                 | FR120, FR118, FR119                                                    |
| Proposition 6.20: Bounded-Pattern Predicates Stabilize Above a Finite Action Bound  | FR215, FR216, FR214                                                    |
| Proposition 6.15: Closure Operations Preserve Exact Certification                   | FR57, FR17, FR18, FR60, FR56, FR59, FR16, FR55, FR58, FR15, FR19, FR20 |
| Proposition 4.2: Decision-Relevant Binary Pairwise Dichotomy                        | FR121, FR123, FR124, FR122                                             |
| Proposition 3.2: Degenerate Positive Cases                                          | FR1, FR6                                                               |
| Proposition 6.1: Exact Certification Depends Only on the Decision Quotient Relation | FR32, FR33, FR31, FR30, FR37, FR51, FR35, FR34                         |
| Proposition 6.19: Bounded Distinct Action Profiles Compress to Bounded Actions      | FR206, FR207, FR205, FR209, FR208                                      |

| Paper claim                                                                           | Lean handle                                          |
|---------------------------------------------------------------------------------------|------------------------------------------------------|
| Proposition 6.9: Duplicate Actions and Duplicate States Preserve Certification        | FR71, FR68, FR54, FR70, FR56, FR69, FR55, FR73, FR72 |
| Proposition 6.7: Explicit Enumeration Is Parameter-Dependent, Not Structural          | FR66, FR67                                           |
| Proposition 2.4: Explicit Primitive Basis                                             | FR29                                                 |
| Proposition 2.2: Explicit Inventory                                                   | FR26, FR25, FR3, FR28, FR27                          |
| Proposition 2.5: All Independent Core Mechanisms Are Nondegenerate                    | FR108, FR106, FR112, FR101, FR105, FR113             |
| Proposition 2.9: Current Positive Interfaces Factor Through Role Classification       | FR4                                                  |
| Proposition 6.12: Invariance Alone Is Too Weak                                        | FR22, FR21, FR23                                     |
| Proposition 2.6: Weak Tree Ordering Does Not Imply Width One                          | FR115, FR114                                         |
| Proposition 3.1: Lifted Cases Introduce No New Primitive                              | FR5                                                  |
| Proposition 6.14: Unrestricted Predicates Trivialize Exact Characterization           | FR145                                                |
| Proposition 4.12: Orbit-Gap Completeness for Exact Classification                     | (no derived Lean handle found)                       |
| Proposition 4.3: Orbit-Gap Template                                                   | FR185                                                |
| Proposition 2.8: Parent-Tree Structure Gives Width-One Decompositions                 | FR110, FR100, FR111                                  |
| Proposition 6.11: Quotient Size Is Unbounded Under Realizability                      | FR24, FR38                                           |
| Proposition 5.3: The Realized Quotient Is Exactly the Label Range                     | FR39, FR40                                           |
| Proposition 6.10: Relabeling Invariance Is Forced by Exact Certification              | FR17, FR18, FR19, FR20                               |
| Proposition 5.2: Every Equivalence Relation Is Optimizer-Realizable                   | FR47, FR46, FR45                                     |
| Proposition 6.13: Independent Summary Frameworks Converge on the Same Structural Core | FR178, FR181, FR182, FR180, FR179                    |
| Proposition 2.7: Cyclic Dependencies Recover Hardness                                 | FR110, FR100, FR111, FR115                           |
| Proposition 6.5: Zero-Distortion Summaries Refine the Optimizer Quotient              | FR177, FR176                                         |
| Theorem 4.8: Admissible Collapse Across the Four Obstruction Families                 | FR190                                                |
| Theorem 4.9: Closure-Sound Package No-Go                                              | FR193                                                |
| Definition 6.17: Admissible Normalization Predicate                                   | FR197, FR199, FR198                                  |
| Theorem 2.1: Explicit Partition of the Current Positive Landscape                     | FR26, FR25, FR1, FR2, FR28, FR27                     |
| Theorem 4.4: Dominant-Pair Admissible No-Go                                           | FR186                                                |
| Theorem 4.6: Ghost-Action Admissible No-Go                                            | FR188                                                |
| Theorem 5.1: Every Labeling Kernel Is Optimizer-Realizable                            | FR7, FR8                                             |
| Theorem 4.5: Margin-Masking Admissible No-Go                                          | FR187                                                |
| Theorem 4.7: Offset Admissible No-Go                                                  | FR189                                                |

Notes: (1) Full rows come from theorem-local inline anchors in this paper. (2) Derived rows are filled by dependency/scaffold claim-handle derivation (same paper-handle label across proof dependencies). (3) Unmapped means no local anchor and no derivable dependency support were found.

Auto summary: mapped 45/47 (full=45, derived=0, unmapped=2).